

ICCAD 2026 #707 — Supplementary Results for Author Response

Measured Layer Sensitivity Guides Mixed-Precision ADC Allocation for CIM-Based LLM Inference

These results support the rebuttal only; the paper is unchanged. Unless noted, the setup is OPT-125M, WikiText-2, p99-clip ADC calibration, paper data split, INT8 per-channel weights. “ Δ PPL/layer” is the per-layer measured sensitivity when a group’s ADC bit-width is reduced. We report *denoised, fixed-seed* measurements; *absolute* Δ PPL magnitudes depend on the calibration protocol and the number of profiling batches, but the allocation-relevant quantities — that W_{fc2} is the most ADC-sensitive group and that the FFN aggregate exceeds the attention aggregate — are reproduced. We report group/aggregate quantities; the single-layer W_{head} group is excluded as numerically unstable (the paper likewise reports the most-sensitive *non- W_{head}* layer, Sec. 5.4).

Table S1. Metric invariance (PPL vs NLL) and weight-quant robustness (OPT-125M, INT8 per-channel weights, probe 7→6). The FFN aggregate exceeds the attention aggregate, and W_{fc2} is the most-sensitive group, under *both* PPL- and NLL-based sensitivity. Because $PPL = \exp(NLL)$ is strictly monotonic with a shared baseline, the full per-group ranking is provably metric-independent: empirically the PPL and NLL rankings are identical (Spearman $\rho = 1.0$) and the group-level ILP allocation is unchanged. (Answers #707C-Q1; supports CR-2 / #707C-C7 / #707D-Q1: the ordering survives a stronger weight-quant floor.)

Sensitivity metric	FFN aggregate	Attention aggregate
Δ PPL/layer (PPL-based)	+0.070	+0.030
Δ NLL/layer ($\times 10^{-4}$, NLL-based)	+2.13	+0.93
W_{fc2} is the single most-sensitive group under both metrics.		
PPL vs NLL ranking: Spearman $\rho = 1.00$; group-level ILP allocation identical.		
Values use the fixed paper-split seed; Table S2 row 1 reports the 3-seed mean.		

Table S2. Robustness of the FFN > attention ordering (aggregate FFN vs aggregate attention mean Δ PPL/layer; INT8 per-channel weights). Swept over probe depth and calibration seed on OPT-125M. FFN > attention holds in 7/9 single-seed OPT settings; seed-averaged it holds clearly at 7→6 (+0.044 vs +0.023) and overwhelmingly at 7→4 ($\approx 11\times$); with a larger profiling budget (calib 16, eval 60) the 7→6 ordering is stable across seeds, indicating the small-probe single-seed noise is a finite-batch artifact. (Supports CR-1/CR-3; answers #707A-W1, #707C-Q2, #707D-W3.)

Model	Probe	FFN Δ PPL/l	Attn Δ PPL/l	FFN > Attn
OPT-125M	7→6 (seed mean)	+0.044	+0.023	yes
OPT-125M	7→6 (3 seeds)	2/3 single-seed; 2/2 at calib16/eval60		2/3
OPT-125M	7→5 (3 seeds)	noisy		2/3
OPT-125M	7→4 (3 seeds)	+0.67 (mean)	+0.063 (mean)	3/3
OPT overall 7/9; clear at 7→6 seed-mean and 7→4.				

Table S3. Full profiling→ILP→PPA pipeline on Qwen2-7B at a usable operating point (WikiText-2; *max-clip* ADC calibration — one of the paper’s three calibration schemes (Table I) — 8-bit nominal; baseline PPL ≈ 16 vs clean ≈ 10). Profiling-guided ILP allocates ADC bits from the measured group sensitivities under an ADC-area budget; ADC energy follows the same 2^b model as in the paper. FFN is far more ADC-sensitive than attention, so the ILP protects the sensitive group and reduces the insensitive high-column-count group. Downstream accuracy (PIQA/BoolQ) is preserved within ≈ 1.6 pp of the CIM-8b baseline. (Answers #707C-C8 and #707A-W2: a complete modern-architecture pipeline at a realistic operating point.)

Configuration	ADC area+energy		PPL	PIQA	BoolQ
Clean (fp16, no CIM)	—		≈ 10	80.4	85.2
CIM uniform 8b (baseline)	—		15.6	77.4	75.0
Profiling-ILP (34% budget)	34%	15.9 (+1.8%)	77.2	73.4	
Profiling-ILP (52% budget)	52%	16.4 (+4.9%)	—	—	—
Proxy-blind (reduce the <i>most</i> -sensitive group): PPL $\rightarrow$ 1938 (model destroyed).					

Analytical mechanism of the inversion (answers #707A-W1, #707C-C3/Q2, #707D-W3). A uniform b -bit ADC with calibrated full-scale V_{FS} splits its error on a MAC output y into a *granular* (resolution) term and an *overload/clipping* term. With saturation rate $s = \Pr[|y| > V_{FS}]$ and step $\Delta = 2V_{FS}/2^b$,

$$D_{\text{ADC}}(b) = \underbrace{D_{\text{gran}}(b)}_{\approx (1-s) \Delta^2/12 \propto (1-s) V_{FS}^2 4^{-b}} + \underbrace{D_{\text{ovl}}}_{\mathbb{E}[(|y|-V_{FS})^2 | |y| > V_{FS}] \text{ (independent of } b)}}.$$

Reducing one bit thus multiplies *only* the granular term by 4 and leaves clipping unchanged. To first order the loss impact of reducing group g is sensitivity $_g \approx \underbrace{\Delta D_{\text{gran},g}}_{\text{granular budget}} \times \underbrace{J_g}_{\text{residual-stream propagation gain}},$ where J_g

is the gain with which an output perturbation of group g reaches the loss along the attention/FFN/residual paths (Fig. 5). The saturation rate captures only the $(1-s)$ part of the granular budget (it ignores V_{FS} and propagation); the Hessian trace approximates propagation but ignores the budget — so each standard proxy sees at most one factor and mispredicts the product, which is exactly why the paper measures ΔPPL directly.

Table S4. Mechanism diagnostics confirm the decomposition (OPT-125M, INT8 per-channel weights, p99-clip, group probe 7→6). This table isolates the two *mechanistic factors*; the sensitivity ranking itself (FFN>attention; W_{fc2} most sensitive at the paper’s operating point) is in Tables S1–S2 and paper Table 4. *Decomposition law*: in every group one-bit reduction scales the granular error D_{gran} by $4.06 \approx 4^1$ and the clipping error D_{ovl} by 1.00 — the bit-width touches only the granular term. *Two anti-correlated factors*: the local granular budget ΔD_{gran} and the propagation gain J are **perfectly anti-correlated across groups** (Spearman $\rho = -1.0$); W_{qkv} receives the *largest* bit-reduction error yet propagates it $\approx 400\times$ more weakly than W_{fc2}/W_{out} (J is hidden-state drift per unit local error; drift and output KL co-move, $\rho = 1.0$) — the quantitative form of the Fig. 5 attenuation argument and the reason high-saturation attention layers are the *least* ADC-sensitive. All diagnostics requested by #707C-C3 (granular/clipping MSE, SQNR, hidden-state drift, output KL) are reported. The *product* of the two factors reproduces FFN>attention; which FFN sub-layer ranks top is set by this operating-point/architecture-dependent product (W_{fc2} at the paper’s max-clip point, W_{fc1} on Qwen2’s gated MLP), explaining the model dependence noted in #707C-W2.

Group	κ (kurt.) heavy tail	$D_{\text{gran}}\uparrow$ (7→6)	$D_{\text{ovl}}\uparrow$ (7→6)	ΔD_{gran} ($\times 10^{-4}$)	SQNR (dB)	KL ($\times 10^{-3}$)	J ($\times 10^3$)
W_{qkv} (attn in)	1.1	4.06	1.00	1.97	27.5	2.82	0.22
W_{out} (attn out)	22	4.06	1.00	0.003	10.8	1.86	127.9
W_{fc1} (FFN up)	47	4.06	1.00	0.256	15.3	4.14	2.09
W_{fc2} (FFN down)	1.5×10^5	4.06	1.00	0.004	4.3	1.84	94.2
Granular budget ΔD_{gran} and propagation gain J are anti-correlated across groups (Spearman $\rho = -1.0$). Sensitivity = their <i>product</i> ; neither saturation ($\rho = -0.70$) nor Hessian ($\rho = 0.20$) captures both factors.							